KSE

Linear algebra basics

2026, Spring Trimester

Course faculty	Mykhailo Bulatetskyi e-mail: mbulatetskyi@kse.org.ua
Course assistants	Antonii Melnyk e-mail: amelnyk002@kse.org.ua Khrystyna Bocharska e-mail: kbocharska@kse.org.ua Ihor Bahrii e-mail: ibahrii@kse.org.ua Polina Khrulyova e-mail: pkhrulyova@kse.org.ua Denys Bezkrovnyi e-mail: dbezkrovnyi@kse.org.ua

Department	Department of Computer Science
Study program	SE29, AI29, CS29, BE29, EBD29
ECTS credits	3 (90 hours)
Class hours	30 hours of problem solving sessions
Course language	Ukrainian
Course format	Offline
Form of study	Full-time studies

Overview

Introduction

Linear algebra is a language of many modern quantitative fields and is overall a certain core in many areas where math is applied. We, however, will learn the basics of linear algebra, that will give you a basis to move forward. In this course, we are going to cover how to work with vectors and matrices and see how they help us organize information, solve systems of equations and describe simple geometric transformations.

You don't have to be a mathematician to be interested in linear algebra, the ideas from this course will keep coming back in statistics, econometrics, optimization, machine learning and many other subjects. Think of it as building a toolkit, part of your skill box you'll use again and again.

Prerequisites

You do not need advanced mathematics to join this course.

If you are reasonably comfortable with high-school algebra - rearranging equations, working with numbers in tables, reading simple graphs and using basic functions - you are ready.

This course is especially useful to you as preparation for probability, statistics, econometrics, optimization and machine learning. We will dive into a linear algebra "grammar", so there is no need to worry about not being ready.

Background and course rationale

Why are we spending time on linear algebra? Because any time you see "many variables changing at once", linear algebra is in the background.

Vectors and matrices give us a compact way to describe real problems:

- systems of equations in economics
- data sets in statistics and machine learning
- simple geometric transformations such as reflections, rotations and scaling

Throughout the course we will work with linear equations, matrices, determinants, inverse matrices and basic geometric transformations. The goal is not just to memorize procedures, but I want you to get used to thinking in terms of linear models so that later quantitative courses feel more natural and less mysterious.

Course aims

By the end of the course, my main aims are that you will:

- feel at home with the basic concepts and techniques of introductory linear algebra
- be able to set up and solve systems of linear equations using matrix methods, not just ad-hoc tricks
- carry out standard operations with vectors and matrices and understand what these operations mean geometrically
- use matrices and systems of equations to model simple real-life situations in economics and other fields
- be well prepared for further courses in mathematics, statistics, econometrics and related areas

Learning outcomes

By the end of the course, you should be able to:

- **transform a matrix to row-echelon form using the Jordan–Gauss method**, and see how this helps solve systems of equations
- **model simple real-life situations with vectors and/or matrices**, for example small economic or data-analysis problems
- **perform basic operations with vectors and matrices** - addition, scalar multiplication, transposition and matrix multiplication - with confidence
- **find the determinant** of a square matrix and interpret what determinant can tell you, what information you can get out of it
- **find an inverse matrix** using the Jordan-Gauss method
- **analyze and solve systems of linear equations (SLE)** using different strategies
- **use matrix transformations to describe basic geometric transformations in the plane** such as symmetry and rotation

Methods of Teaching and Assessment

Before class you'll watch brief video lectures where new ideas are introduced with simple examples. You can pause, rewind and rewatch them whenever you need, so you come to class already knowing the basic story.

In class we'll focus on your questions: clearing up confusing points, going over tricky steps and connecting the theory to specific problems.

A big part of the course is working through problems together - individually, in pairs and in small groups - so you use the concepts, see typical exam-style tasks and get feedback on your thinking. That's why I encourage you to spend time in a group of your fellow students and try to crack a hard problem or two.

To make it short for you:
- Video lectures for self-study of new concepts.
- Problem solving sessions devoted to working through exercises and applications.

Regular homework sets where you don't just give the final answer but also write out your reasoning step by step. This is your chance to practice, make mistakes, and learn through problem-solving.

From time to time, we'll have checks in class focused on specific topics. They're there to see what you've really understood, not to surprise you - questions will look very similar to what you've seen in videos and homework.

Your activity during classes also matters: asking questions, suggesting ideas, and sometimes going to the board to explain your idea of a solution. The goal is to reward genuine engagement, not just right answers.

Again, to make it simple for you:
- Homework with written solutions and detailed explanations.
- In-class tests checking understanding of specific topics.
- Participation in problem solving sessions (activity, engagement, work at the board).

Participation is not graded as a separate component.

Course Structure

Course duration: 7 weeks (within the 2026 academic year, from **05.01 to 22.02**).

Credits and hours: 3 ECTS (90 hours total workload), including 30 hours of problem solving sessions and approximately 60 hours of independent study. Below is the structure of instructional and assessment methods used in this course:

- Problem solving sessions (15 sessions): intensive practice in small groups or whole class format; solving and discussing problems at the board.

- Flipped lessons: students are expected to watch video lectures and read basic material before problem solving sessions.

Tests:

- 3 tests in total, held in weeks 3, 5 and 7.

- Each test lasts up to 60 minutes and includes several open problems and (possibly) multiple-choice questions.

- Students may bring one A4 one-sided summary sheet; no other materials or devices are allowed.

Homework tasks:

- 3 homework sets (weeks 1, 3 and 5).

- Each homework must be submitted as a single PDF file (portrait orientation, pages in the correct order). The file name should contain the student's last and first name.

- Late submissions ***are not accepted**** and are graded with 0 points.

Except for the one allowed late day described below*

Grading Policy

Written assignments (homework tasks, quizzes and tests) grading. Taking into account the fact that the course is about mathematics and studying at the university involves observing the principle of scientificity, all answers and conclusions should be justified. In addition, a detailed explanation of the process of solving the problem is a good practice that allows a student to appeal in case he/she does not agree with evaluation of his/her work. Solutions or answers without sufficient explanation may receive a reduced or zero grade.

Maximum number of points: 107 points in total (the final course grade is capped at 100 points and cannot be negative).

Components:

- Quizzes: 32 points total (4 × 8 points). Quizzes can be taken in any week during the course.

- Homework assignments : 15 points total (3 × 5 points). Homework tasks are given in weeks 1, 3 and 5.

- Tests: 60 points total (3 × 20 points). Tests will take place in weeks 3, 5 and 7.

Homework problems and due time are announced on the course page in Moodle. All homework solutions should be written legible using dark ink on a white paper, photographed straightly without rotation, combined in order to a single pdf file, and submitted to Moodle before the announced deadline. Filename should contain last and first name of executant. Submissions not complying with all rules above will not be accepted or graded at the teacher's discretion. In this case, the student receives zero points for the assignment (no resubmission allowed). We will have a late day policy on homework tasks. ***Each student has one late day, i.e., you may submit your homework up to 24 hours after the deadline****. Always reach out if you're facing unusual disruptions to your classwork. You are allowed to work on the homework in small groups, but you must write up your own homework to hand in. Do not copy solutions from people you have worked with or from anyone else. If the answer is given in a book, don’t just copy it, explain how you got it. Always indicate if you used a calculator for calculations.

Grade of the course is calculated as a sum of all the activities from the table above but not more than 100 points.

General Grading Scheme

Task	***Points per activity***	***Total points***
Quiz 1, 2, 3, 4	***8 points***	***32 points***
Homework 1, 2, 3	***5 points***	***15 points***
Test 1, 2, 3	***20 points***	***60 points***
Total score:		***107 points***

Academic integrity warning

Academic integrity is highly valued by the KSE community and its vast majority of students. We do not tolerate plagiarism and cheating. Very competitive essence of the program calls for absolutely equal treatment of all students. Even a single case of violation of academic integrity is a serious misdemeanor which may lead to unjust redistribution of grades and, consequently, overall rankings, possible grants, fee reductions and other merit-based awards.

The minimum penalty for a student caught cheating or plagiarizing during any practical task will be a zero for this assignment with the consecutive decrease of the final course grade by at least one letter-grade. Repetitive violation of academic integrity may lead to the dismissal of the student from KSE. By enrolling on this course, students affirm that they are familiar with the KSE Code of Academic Integrity.

AI Tools usage policy

You're welcome to use AI tools to help you **study**. Just keep in mind: they should help you learn, **not do the assignment instead of you**. In this course you're expected to show your own reasoning and computations, so AI is fine for explanations and quick checks, but your final submission must be **your own work**.

What **is allowed**: you can use AI to clarify concepts in plain language (for example, to understand a definition), to support studying (for example, to generate extra practice questions that are not your graded homework) and to help you reflect on your own work.

What **is not allowed**: you may not use AI to generate answers or solution paths for graded work (homework, quizzes, tests) in a way that **replaces your own effort**. For example, this includes asking for a complete solution or next step of solution, when you are stuck, or requesting a full explanation and then submitting it as if it were your own. You also may not use AI to bypass the main learning goals by outsourcing core reasoning or computations (for example, having AI solve systems, compute determinant, etc.) and then copying the result and/or lightly rewriting it.

To keep things fair for everyone, for graded work you should show **enough intermediate steps** so that your **reasoning is clear**. A correct final answer without supporting work may receive **reduced credit**. If you do use AI while studying, a short note like "**used AI to clarify a definition**" or "**used AI to check my arithmetic**" is appreciated (no need to share prompts). In some cases, your teacher may ask you to briefly explain your submitted work, but this is not meant to be stressful – just a simple way to **confirm your understanding**.

If a submission appears to rely on AI in a way that violates this policy, it may receive **no credit** and you may be asked to redo the work or complete an alternative assessment, in line with the course and KSE **academic integrity rules**. Finally, this AI tools policy **may be updated during the course**, any changes will be communicated.

Support for Students in Active Military Service

The course is open to students who are currently on active military service. We understand that service obligations may create unpredictable constraints. If this applies to you, please inform the teacher as early as possible. We will work with you individually to create a feasible study plan while preserving academic standards and learning outcomes. Possible accommodations may include:

- flexible deadlines
- alternative formats for participation
- rescheduling quizzes/tests or providing an equivalent assessment
- a personalized sequence of topics and checkpoints adapted to your availability. Students are not required to disclose sensitive details beyond what is necessary to arrange academic accommodations.

Course Plan

Difficulty key: ● Basic ◆ Core ▲ Advanced

Schedule

Week 1 (Jan 5 – Jan 11, 2026) ● Basic

Topic 1: Linear equations. Systems of linear equations (SLE). Matrices of SLE. Jordan–Gauss method.

6 class hours (problem solving: 6h)	Activity	Content
	Introduction lecture	• Answering questions about course structure/organization
2h problem solving	Problem Solving Session 1. Back to School.	• System of linear equations (SLE)
2h problem solving	Problem Solving Session 2. Matrices and row operations.	• Matrices of SLE, row operations
2h problem solving	Problem Solving Session 3. Jordan-Gauss method elimination.	• Gaussian elimination

Week 2 (Jan 12 – Jan 18, 2026) ◆ Core **Topic 1: Linear equations. Systems of linear equations (SLE). Matrices of SLE. Jordan–Gauss method.** **Topic 2: Vectors, matrices and operations with them.**		
6 class hours (problem solving 6h)	Activity	Content
2h problem solving	Problem Solving Session 4. Applications for SLE.	• Applications for solutions for SLE
2h problem solving	Problem Solving Session 5. Test Preparation.	• Preparation for Test 1
2h problem solving	Problem Solving Session 6. Operations with vectors. The concept of a linear space.	• Algebraic view on vectors, definition of a matrix (recap), special matrices

Week 3 (Jan 19 – Jan 25, 2026) ◆ Core **Topic 2: Vectors, matrices and operations with them.**		
4 class hours (problem solving 4h)	Activity	Content
2h problem solving 2h problem solving	Problem Solving Session 7. Matrix operations: addition, scalar multiplication, transposition. Problem Solving Session 8. Matrix operations: multiplication. Test 1	• Operations with matrices: addition, scalar multiplication, transposition • Multiplication of matrix by vector, matrix multiplication

Week 4 (Jan 26 – Feb 1, 2026) ▲ Advanced

Topic 2: Vectors, matrices and operations with them.

Topic 3: Determinant of a matrix, inverse matrices.

4 class hours (problem solving 4h)	Activity	Content
2h problem solving	Problem Solving Session 9. Test preparation.	• Preparation for Test 2
2h problem solving	Problem Solving Session 10. Finding an inverse matrix.	• Inverse matrix. Finding inverse matrix using Jordan-Gauss method. Explicit formula for an inverse for 2×2 matrix

Week 5 (Feb 2 – Feb 8, 2026) ▲ Advanced **Topic 3: Determinant of a matrix, inverse matrices**		
4 class hours (problem solving 4h)	Activity	Content
2h problem solving 2h problem solving	Problem Solving Session 11. Solving SLE using the inverse matrix. Problem Solving Session 12. Determinants and their applications. Test 2	• Solving SLE using the inverse matrix • Determinant of matrix. Geometrical approach in the case of a 2×2 matrix

Week 6 (Feb 9 – Feb 15, 2026) ▲ Advanced

Topic 3: Determinant of a matrix, inverse matrices

Topic 4: Geometric transformations using matrix operations. Course wrap-up.

4 class hours (problem solving 4h)	Activity	Content
2h problem solving	Problem Solving Session 13. Test preparation.	• Preparation for Test 3
2h problem solving	Problem Solving Session 14. Geometric transformations and matrices.	• Geometric transformations using vectors/matrices: symmetry and rotation

Week 7 (Feb 16 – Feb 22, 2026) ● Basic **Topic 4: Geometric transformations using matrix operations. Course wrap-up.**		
2 class hours (problem solving 2h)	Activity	Content
2h problem solving	Problem Solving Session 15. Total recall Test 3	• Course wrap-up